

Notes on Heston Stochastic Volatility Model and Its Use in Option Pricing

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1 Introduction

Suppose we would like to find the expected payoff for a European call option written on an underlying asset S_t , where $t \in [0, T]$, with strike price K and maturity T . The payoff of this option at time T is

$$\text{Payoff} = (S_T - K)_+ = \max(S_T - K, 0).$$

This payoff structure reflects the fact that a call option is exercised only when the terminal stock price exceeds the strike price.

Under the risk-neutral pricing framework, the expected present value of the option can be written as:

$$C_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)_+],$$

where r is the continuously-compounded risk-free interest rate and $\mathbb{E}^{\mathbb{Q}}[\cdot]$ denotes expectation under the risk-neutral measure $\mathbb{Q}$. The reason the expected payoff is important is that, under risk-neutral valuation, the option premium can be determined from the expected payoff when discounted.

There are a variety of pricing methodologies. Discrete methods such as binomial trees, and methods assuming continuous price movements such as the geometric Brownian motion are used. These methods are motivated by the assumption of constant volatility in time. For the sake of simplicity, we assume:

$$dS_t = rS_t dt + \sigma S_t dW_t,$$

where $\sigma > 0$ is constant and W_t is a standard Brownian motion. This is one of the few stochastic differential equations with a closed-form solution, and the method for pricing is relatively straightforward. However, constant volatility is unrealistic. In real markets, volatility changes through time and is not properly modeled in a constant-volatility system.

As such, we must introduce stochastic models which assume time-dependent volatility terms. Since closed-form solutions under these assumptions would be difficult to find, we may use Monte Carlo simulation to simulate many sample

paths of S_t , compute discounted payoffs, and average them. In this project, we simulate S_t using the Heston stochastic volatility model, compute a Monte Carlo premium estimate, and study the variance and confidence intervals of this estimator. We then apply a control variate method to reduce the variance and improve computational efficiency

2 The Heston Stochastic Volatility Model

The Heston model assumes the asset price S_t and variance v_t satisfy the following:

$$dS_t = rS_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \quad (1)$$

$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^{(2)}, \quad (2)$$

where $\kappa > 0$ is the mean reversion speed, $\theta > 0$ is the long-run variance level, and $\xi > 0$ is the volatility-of-volatility.

The Brownian motions are correlated:

$$dW_t^{(1)} dW_t^{(2)} = \rho dt, \quad -1 \leq \rho \leq 1.$$

Intuitively, we assume that the Brownian motions for both systems in the model are correlated because price movements have an inherent effect on volatility. In a healthy market we may observe that when the stock price decreases, volatility tends to increase. This corresponds to the typical assumption $\rho < 0$. However, this is not always the case, and ρ can vary depending on the asset class and the nature of the market.

3 Monte Carlo Discretization

Let M be the number of time steps and $\Delta t = T/M$. We generate a pair of correlated standard normals (Z_1, Z_2) with correlation ρ at each step. Below are the discretized equations for the system. We assume that v_0 and S_0 are given.

$$v_{t+\Delta t} \approx v_t + \kappa(\theta - v_t)\Delta t + \xi \sqrt{v_t} \sqrt{\Delta t} Z_2,$$

$$S_{t+\Delta t} \approx S_t \exp\left(\left(r - \frac{1}{2}v_t\right) \Delta t + \sqrt{v_t} \sqrt{\Delta t} Z_1\right).$$

Parameter Values Used

Below you can see my graph of the simulated Heston paths for a sample asset (Figure 1), where the parameters used were:

$$\begin{aligned} S_0 &= 200, \\ T &= 1.0, \\ r &= 0.04, \\ \text{steps} &= 252, \\ \kappa &= 2.0, \\ \theta &= 0.04, \\ v_0 &= 0.04, \\ \rho &= -0.473, \\ \xi &= 0.6, \\ N_{\text{paths}} &= 6. \end{aligned}$$

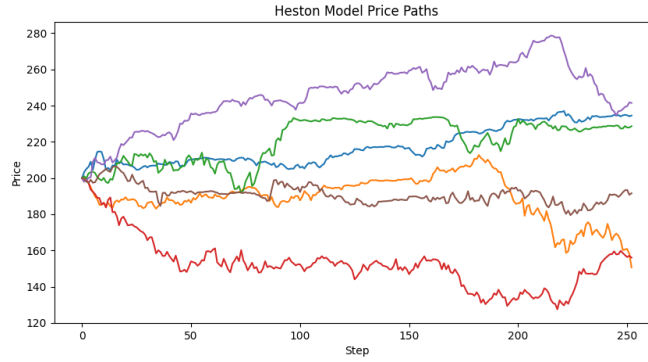


Figure 1: Simulated Heston stock price paths using the chosen parameter values.

Monte Carlo Pricing Procedure

The Monte Carlo pricing procedure used in this project can be summarized as follows:

1. Choose parameters $(S_0, v_0, r, \kappa, \theta, \xi, \rho)$ and a time grid with M steps.
2. For each path $i = 1, \dots, N$, simulate $(S_t^{(i)}, v_t^{(i)})$ forward in time using the discretized Heston updates.
3. Compute the discounted payoff $X_i = e^{-rT}(S_T^{(i)} - K)_+$.
4. Estimate the call premium using $\hat{C}_0 = \frac{1}{N} \sum_{i=1}^N X_i$.

In the numerical results below, we price a European call option with strike $K = 210$ and maturity $T = 1$ year.

Discretization Considerations

Since the Heston model is a continuous-time model, the simulation depends on the scheme and the step size Δt . Smaller step sizes generally reduce error but increase computational cost. In this project, I used $M = 252$ steps to represent daily sampling over one year.

4 Variance of the Call Payoff Under Heston

After simulating N paths, let $S_T^{(i)}$ be the terminal price on path i . Define the discounted payoff random variable

$$X = e^{-rT}(S_T - K)_+, \quad \text{so that} \quad X_i = e^{-rT}(S_T^{(i)} - K)_+.$$

Then the Monte Carlo price estimator is

$$\hat{C}_0 = \frac{1}{N} \sum_{i=1}^N X_i.$$

The sample variance of X is

$$\widehat{\text{Var}}(X) = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})^2, \quad \bar{X} = \frac{1}{N} \sum_{i=1}^N X_i,$$

and the variance of the Monte Carlo estimator satisfies

$$\widehat{\text{Var}}(\hat{C}_0) = \frac{\widehat{\text{Var}}(X)}{N}, \quad \text{so } \text{SE}(\hat{C}_0) = \sqrt{\widehat{\text{Var}}(X)/N}.$$

This relationship is important because it explains how Monte Carlo accuracy improves as the number of simulated paths increases. Specifically, the standard error decreases at the rate $1/\sqrt{N}$.

Confidence Interval Interpretation

Using the central limit theorem, an approximate 95% confidence interval for the option price is

$$\hat{C}_0 \pm 1.96 \cdot \text{SE}(\hat{C}_0).$$

Confidence intervals provide a practical measure of Monte Carlo accuracy and allows us to compare the accuracy between different simulations.

Monte Carlo Results Summary

N paths	Premium estimate	$\widehat{\text{Var}}(X)$	$\widehat{\text{Var}}(\widehat{C}_0)$	SE
1,000	\$14.1035	485.7263	0.4857	0.6969
100,000	\$13.5415	437.5771	0.0044	0.0661
1,000,000	\$13.5169	437.6058	0.0004	0.0209

Interpretation of Results

As the number of simulated paths increases, the Monte Carlo premium estimate stabilizes and the standard error decreases, which leads to narrower confidence intervals. This is consistent with the fact that Monte Carlo error decreases at rate $1/\sqrt{N}$. In other words, increasing N improves the accuracy of the estimate, but doing so can be computationally expensive for large simulations.

Control Variate Variance Reduction

Although increasing N reduces estimator variance, it can be computationally expensive. A common technique to improve Monte Carlo efficiency is *variance reduction*. One of the most widely used methods is the control variate method.

To further reduce variance, I introduced a control variate. The idea behind a control variate is to use an additional random variable that is correlated with the payoff and has a known expected value. This allows us to remove some of the randomness in the payoff estimator without changing the true expected value.

I chose the control variate

$$Y^{(c)} := e^{-rT} S_T.$$

Under the risk-neutral framework, the expected value of this random variable is

$$\mathbb{E}[Y^{(c)}] = S_0,$$

since $\mathbb{E}^{\mathbb{Q}}[e^{-rT} S_T] = S_0$.

Define the control variate estimator

$$Z = X + c(Y^{(c)} - \mu_c), \quad \text{where } \mu_c = \mathbb{E}[Y^{(c)}] = S_0.$$

Because $\mathbb{E}[Y^{(c)} - \mu_c] = 0$, we still have $\mathbb{E}[Z] = \mathbb{E}[X] = C_0$. The goal is to choose c so that $\text{Var}(Z)$ is minimized.

We compute

$$\text{Var}(Z) = \text{Var}(X) + 2c \text{Cov}(X, Y^{(c)}) + c^2 \text{Var}(Y^{(c)}).$$

Differentiating with respect to c and setting the derivative equal to zero yields

$$c^* = -\frac{\text{Cov}(X, Y^{(c)})}{\text{Var}(Y^{(c)})}.$$

This coefficient can also be interpreted as the slope from a simple linear regression of X on $Y^{(c)}$ (with an intercept).

Control Variate Results Summary

N paths	Control variate premium	$\widehat{\text{Var}}(Z)$	Variance reduction factor
100,000	\$13.5724	151.8959	2.8808
1,000,000	\$13.5373	150.7753	2.9024

Since the variance reduction factor is approximately 2.9, the control variate estimator achieves roughly the same accuracy as plain Monte Carlo using about 2.9 times as many paths.

Comparing Confidence Intervals

Using $N = 1,000,000$ paths, the computed confidence intervals were:

Without control 95% CI: (\$13.47594, \$13.55794),

Control variate 95% CI: (\$13.51322, \$13.56136).

The width of the confidence interval without the control variate was approximately \$0.0820, while the width with the control variate was approximately \$0.0481. This shows that the control variate produces a noticeably tighter confidence interval. In particular, the confidence interval width shrank by a factor of about 1.70, meaning the control variate estimate is substantially more precise using the same number of simulated paths.

95% Confidence Interval Comparison ($N = 1,000,000$)

Method	95% CI	CI width
Plain Monte Carlo	(\$13.47594, \$13.55794)	\$0.0820
Control Variate	(\$13.51322, \$13.56136)	\$0.0481

5 Limitations and Remarks

The accuracy of Monte Carlo pricing depends on both the number of paths N and the step size Δt . Although increasing N reduces sampling error, error may still remain due to the Euler approximation of the variance process. More advanced schemes (such as the Milstein method or exact simulation approaches) may further improve accuracy. Additionally, the Heston model is parameter-driven, meaning that realistic pricing depends on being able to choose reasonable $(\kappa, \theta, \xi, \rho, v_0)$.

6 Conclusion

In this project, I used the Heston stochastic volatility model to simulate stock price paths and price a European call option under a risk-neutral framework. The Monte Carlo method provided an estimate of the call premium by averaging

discounted payoffs across many simulated paths. I also computed the variance and standard error of the estimator to quantify simulation uncertainty and construct confidence intervals.

Finally, I implemented a control variate using the discounted terminal stock price. This variance reduction technique significantly improved the efficiency of the simulation by reducing the payoff variance and producing tighter confidence intervals. Overall, these results show that Monte Carlo simulation is a flexible method for option pricing under stochastic volatility assumptions, and that variance reduction methods can provide meaningful improvements in accuracy without increasing computational cost.

I would like to continue looking at more continuous-time models, and at variance reduction techniques that may allow for more accurate results at less computational cost. While this paper focused on the Heston Model, there are a variety of models that rely on different assumptions when pricing. What interested me while I was reading was the use of a counting process to describe sudden jumps in the volatility of the value of an asset as an addition to the randomness added by the correlated brownian motions in this model.